Name: Chey Naryvety

ID: IDTB110199

W08

Lab1

```
ashxio@ashio:~/public_html/naryvety.chey/lab08$ mkdir data/dir1 data/dir2
ashxio@ashio:~/public_html/naryvety.chey/lab08$ touch data/file1.txt data/file2.txt data/file3.log
ashxio@ashio:~/public_html/naryvety.chey/lab08$ touch data/dir1/file11.txt data/dir1/file12.log
ashxio@ashio:~/public_html/naryvety.chey/lab08$ touch data/dir2/file21.txt data/dir2/file21.log4
ashxio@ashio:~/public_html/naryvety.chey/lab08$ ls
backups  data  file1.txt  file2.txt  scripts  testmydir  testmyfile.txt
ashxio@ashio:~/public_html/naryvety.chey/lab08$ cd data/
ashxio@ashio:~/public_html/naryvety.chey/lab08/data$ ls
dir1  dir2  file1.txt  file2.txt  file3.log
ashxio@ashio:~/public_html/naryvety.chey/lab08/data$ ls dir1/
file11.txt  file12.log
ashxio@ashio:~/public_html/naryvety.chey/lab08/data$ ls dir2/
file21.log4  file21.txt
ashxio@ashio:~/public_html/naryvety.chey/lab08/data$ |
```

Lab2

```
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ nano print_args.sh
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ chmod +x print_args.sh
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ ./print_args.sh "one" "two three four" five
1st arg is "one"
2nd arg is "two three four"
3rd arg is "five"
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ |
```

```
#!/bin/bash

print_my_args() {
   echo "1st arg is \"$1\""
   echo "2nd arg is \"$2\""
   echo "3rd arg is \"$3\""
}

print_my_args "$@"
```

Lab3

```
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ nano print_all.sh
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ chmod +x print_all.sh
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ ./print_all.sh "one" "two 3 four" five six 7 8 9 10 11 a b c
Argument 1 is "one"
Argument 2 is "two 3 four"
Argument 3 is "five"
Argument 4 is "six"
Argument 5 is "7"
Argument 6 is "8"
Argument 7 is "9"
Argument 8 is "10"
Argument 9 is "11"
Argument 10 is "a"
Argument 11 is "b"
Argument 12 is "c"
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ |
```

```
print_all_my_args() {
   count=1
   for args in "$@"
   do
        echo "Argument $count is \"$args\""
        ((count++))
   done
}
print_all_my_args "$@"
```

Lab4

```
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ cp ../../lab07/scripts/myscript.sh myscript.sh
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ chmod +x myscript.sh
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ ./myscript.sh "one" "../data/file1.txt" -a ../data ../backups/ 2
./ -b
Opt 1 is "-a"
Dir 1 is "../data"
Dir 2 is "../backups/"
Opt 2 is "-b"
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$
```

```
#!/bin/bash

OPT_COUNT=1
DIR_COUNT=1

for ARGS in "$@"
do
   if [[ -d "$ARGS" ]]
   then
       echo "Dir $DIR_COUNT is \"$ARGS\""
       ((DIR_COUNT++))
  elif [[ "$ARGS" == -* ]]
  then
       echo "Opt $OPT_COUNT is \"$ARGS\""
       ((OPT_COUNT++))
  fi
done
```

Assignment W08-1

```
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ cp ../../lab07/scripts/backup_files.sh backup_files.sh
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ cp ../../lab07/scripts/count_files.sh count_files.sh
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ nano backup_files.sh
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ chmod +x backup_files.sh
```

```
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ ./backup_files.sh ../data/ ../backups/
Backup completed successfully!
Files copied from '../data/' to '../backups/'
Number of files in '../data/': 3
Number of files in '../backups/': 3
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ ./backup_files.sh -s
Usage:
 ./backup_files.sh <source_dir> <backup_dir>
 ./backup_files.sh -s <source_dir> -b <backup_dir>
 ./backup_files.sh -b <backup_dir> -s <source_dir>
Examples:
 ././backup_files.sh ../data/ ../backups/
 ././backup_files.sh -s ../data/ -b ../backups/
 ././backup_files.sh -b ../backups/ -s ../data/
Notes:
 - <source_dir> must exist and be a directory
 - <backup_dir> will be created if it does not exist
 - The script copies ALL regular files recursively and preserves subfolders.
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$
```

localhost:8000/naryvety.chey/lab08/scripts/backup_files.sh

```
#!/bin/bash

show_instruction() {
    echo "Usage:"
    echo " $0 <source_dir> <backup_dir>"
    echo " $0 -s <source_dir> -b <backup_dir>"
    echo " $0 -b <backup_dir> -s <source_dir>"
    echo "Examples:"
    echo " ./$0 ../data/ ../backups/"
    echo " ./$0 -s ../data/ -b ../backups/"
    echo " ./$0 -b ../backups/ -s ../data/"
    echo "Notes:"
    echo " - <source_dir> must exist and be a directory"
    echo " - <backup_dir> will be created if it does not exist"
    echo " - The script copies ALL regular files recursively and preserves subfolders."
    exit 1
}
check_args() {
  if [[ $# -eq 2 ]]; then
      SOURCE="$1"
      BACKUP="$2"
  elif [[ $# -eq 4 ]]; then
      if [[ $1 == "-s" && $3 == "-b" ]]; then
          SOURCE="$2"
          BACKUP="$4"
      elif [[ $1 == "-b" && $3 == "-s" ]]; then
          BACKUP="$2"
          SOURCE="$4"
      else
          show_instruction
      fi
  else
      show_instruction
  fi

  if [[ ! -d "$SOURCE" ]]; then
      echo "ERROR: Source directory '$SOURCE' does not exist!"
      exit 1
  fi
}
backup_files() {
  if [[ ! -d "$BACKUP" ]]; then
      mkdir -p "$BACKUP"
  fi

  cp -r "$SOURCE"/* "$BACKUP"

  echo "Backup completed successfully!"
  echo "Files copied from '$SOURCE' to '$BACKUP'"

./count_files.sh "$SOURCE"
./count_files.sh "$BACKUP"
}

check_args "$@"
backup_files
```

Assignment W08-1 (Bonus)

```
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ cp backup_files.sh backup_program.sh
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ nano backup_program.sh
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ chmod +x backup_program.sh
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ ./backup_program.sh ../data/ ../backups/
Backup completed successfully!
Files copied from '../data/' to '../backups/'
Number of files in '../data/': 3
Number of files in '../backups/': 3
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ ls ../backups/
dir1  dir2  file1.txt  file2.txt  file3.log
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ ls ../backups/dir1/
file11.txt  file12.log
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ ls ../backups/dir2/
file21.log4  file21.txt
ashxio@ashio:~/public_html/naryvety.chey/lab08/scripts$ |
```

localhost:8000/naryvety.chey/lab08/scripts/backup_program.sh

```
#!/bin/bash

show_instruction() {
    echo "Usage:"
    echo " $0 <source_dir> <backup_dir>"
    echo " $0 -s <source_dir> -b <backup_dir>"
    echo " $0 -b <backup_dir> -s <source_dir>"
    echo "Examples:"
    echo " ./$0 ../data/ ../backups/"
    echo " ./$0 -s ../data/ -b ../backups/"
    echo " ./$0 -b ../backups/ -s ../data/"
    echo "Notes:"
    echo " - <source_dir> must exist and be a directory"
    echo " - <backup_dir> will be created if it does not exist"
    echo " - The script copies ALL regular files recursively and preserves subfolders."
    exit 1
}
check_args() {
  if [[ $# -eq 2 ]]; then
      SOURCE="$1"
      BACKUP="$2"
  elif [[ $# -eq 4 ]]; then
      if [[ $1 == "-s" && $3 == "-b" ]]; then
          SOURCE="$2"
          BACKUP="$4"
      elif [[ $1 == "-b" && $3 == "-s" ]]; then
          BACKUP="$2"
          SOURCE="$4"
      else
          show_instruction
      fi
  else
      show_instruction
  fi

  if [[ ! -d "$SOURCE" ]]; then
      echo "ERROR: Source directory '$SOURCE' does not exist!"
      exit 1
  fi
}
backup_files() {
  if [[ ! -d "$BACKUP" ]]; then
      mkdir -p "$BACKUP"
  fi

  cp -r "$SOURCE"/. "$BACKUP"

  echo "Backup completed successfully!"
  echo "Files copied from '$SOURCE' to '$BACKUP'"

./count_files.sh "$SOURCE"
./count_files.sh "$BACKUP"
}

check_args "$@"
backup_files
```